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1 The proof of the collar neighbourhood theorem for smooth manifolds

1.1 Prerequisites

1.1.1 Vector fields

Lemma 1.1.1. *Let M be a smooth n-manifold with boundary. Let B be a coordinate half ball with corresponding map $\varphi\colon B \to \mathbb{H}^n$. Then $\frac{\partial}{\partial x^n}$ is a smooth vector field on B which is inward-pointing when restricted to $\partial M \cap B$.*

Proof. The vector field is smooth by Example 8.2 in [Lee06]. Let $p \in B$. We need to show that $\frac{\partial}{\partial x^n}\Big|_p$ is inward-pointing. By Proposition 5.42 in [Lee06], it suffices to show that the n-th component of $\frac{\partial}{\partial x^n}\Big|_p$ is positive. Its n-th component is 1 so we are done. □

Lemma 1.1.2. *Let M be a smooth manifold, A a closed and U an open subset of M with $A \subseteq U$. Let $X\colon A \to TM$ be a smooth vector field along A. Then there is a smooth vector field $\tilde{X}\colon M \to TM$ with support in U which agrees with X on A.*

Proof. We first notice that the proof of Lemma 2.26 (Extension Lemma for Smooth Functions) in [Lee06] also works for the domain TM . Now we need to verify that a smooth map of the form

General Lemma?

$$\tilde{X} = \sum_{p \in A} \tau_p(x)\tilde{X}_p(x)$$

is a vector field. Here the $\tau_p x$ are bump functions and the $\tilde{X}_p$ local extensions of the vector field. Now locally this sum is finite. Module over $C^\infty(M)$ ... blah blah blah □

Lemma 1.1.3 (Problem 8-4 in [Lee06])**.** *Let M be a smooth n-manifold with boundary. There exists a smooth vector field on M, which is inward-pointing when restricted to the boundary.*

Proof. Cover ∂M with a countable number of regular coordinate half-balls $(B_i)_i$. This is possible by Exercise 1.42 in [Lee06]. Let us call the corresponding larger coordinate half-balls B_i' and the corresponding smooth coordinate map $\varphi_i\colon B_i' \to \mathbb{H}^n$. By Lemma 1.1.1, $\frac{\partial}{\partial x^n}$ is a smooth vector field on B_i' which is inward-pointing when restricted to $\partial M \cap B$. In particular, $\frac{\partial}{\partial x^n}$ is a smooth vector field along B_i. Using Lemma 1.1.2, we can extend these to smooth vector fields $X_i\colon M \to TM$ with support in B_i' and which agree with $\frac{\partial}{\partial x^n}$ on B_i. This is again inward-pointing which is easy to see using Proposition 5.42 in [Lee06] . Now take a partition of unity subordinate to the cover $\{B_i \mid i\} \cup M \setminus \partial M$. Then do the same thing as in the proof of Lemma 1.1.2 and get a global smooth vector field that is inward-pointing on the boundary. □

write Lemma

1.1.2 Flows and Flowouts

Lemma 1.1.4. *Let U be an open subset of $\mathbb{H}^n$ that has non-zero intersection with the boundary. Let V be a smooth vector field on U. Then there exists a smooth map $\delta_U : U \to \mathbb{R}+$ and a smooth open embedding $\Phi_U :$*

Lemma 1.1.5. *Let M be a smooth n-manifold with boundary. Let N be a smooth vector field on M that is inward-pointing when restricted to the boundary. Let (U, φ) be a boundary chart. Then there exists a smooth map $\delta_U : U \to \mathbb{R}^+$ and a smooth open embedding $\Phi_U : P_U \to M$ where $P_U =:= \{(t, p) \in \mathbb{R} \times U \mid 0 \leq t < \delta_U(p)\}$ ...*

Proof. Consider the pushforward $\varphi_* X$ of X by φ on $\varphi(U) \subseteq \mathbb{H}^n$. Extend $\varphi_* X$ to a vector field V on an open subset U' of $\mathbb{R}^n$ with $U' \cap \mathbb{H}^n = U$ such that still $V^n(p) > 0$ for all $p \in \partial\mathbb{H}^n$. Now let $\theta_V : \mathcal{D} \to U'$ be the flow of V as given by the Fundamental Theorem of Flows (Theorem 9.12 in [Lee06]) where $\mathcal{D} \subseteq \mathbb{R} \times M$ is a flow domain. Since $\varphi(U) \cap \partial\mathbb{H}^n$ is an embedded $n-1$-dimensional submanifold of U' and V is clearly nowhere tangent to it, we can apply the Flowout Theorem (Theorem 9.20 in [Lee06]) to get a smooth function $\delta'_U : \varphi(U) \cap \partial\mathbb{H}^n \to \mathbb{R}^+$ such that $\theta_V|_{\mathcal{O}} : \mathcal{O} \to U$ is a smooth embedding where $\mathcal{O} := \{(t, p) \in (\mathbb{R} \times (\varphi(U) \cap \partial\mathbb{H}^n) \cap \mathcal{D}) \mid 0 \leq < \delta'_U(p)\}$.

How? -> Lemma

show $t \geq 0$

Now define □

Lemma 1.1.6 (Boundary Flowout Theorem, Theorem 9.24 and Problem 9-11 in [Lee06])**.** *Let M be a smooth manifold with non-empty boundary. Let N be a smooth vector field on M that is inward-pointing when restricted to the boundary. Then there exists a smooth map $\delta : \partial M \to \mathbb{R}^+$ and a smooth embedding $\Phi_\delta : P_\delta \to M$ where $P_\delta := \{(t, p) \in \mathbb{R} \times \partial M \mid 0 \leq t < \delta(p)\}$ such that $\Phi_\delta(P_\delta)$ is a neighbourhood of ∂M and for every $p \in \delta M$, $t \mapsto \Phi_\delta(t, p)$ is an integral curve of N starting at p.*

Proof. Cover ∂M with a countable number of boundary charts (U, φ). □

2 Inward Pointing Vectors in Manifolds with Corners

Defining the notion of inward pointing vectors in manifolds with corners is substantially harder than in manifolds with boundary without corners. Let us first examine why this is.

Here is a common definition for manifolds with boundary without corners:

Definition 2.0.1 ([Lee06] p. 136)**.** Let M be a smooth manifold with boundary without corners. Let $p \in M$. Then $v \in T_pM \setminus T_p\partial M$ is inward pointing if and only if there is an $\varepsilon > 0$ and a smooth curve $\gamma\colon [0, \varepsilon) \to M$ such that $\gamma(0) = p$ and $\gamma'(0) = v$.

This definition cannot be used for manifolds with corners as in this case ∂M is not always a manifold and thus $T_p\partial M$ is not defined.

The existence of the $\gamma\colon [0, \varepsilon) \to M$ by itself characterises whether a vector is inward pointing or tangent to the boundary, a notion which we will call *realisable* or *weakly inward pointing*.

So for manifolds with corners we either need a completely new definition that intrinsically excludes the vectors tangent to the boundary or we need a definition of realisability and a way to exclude the tangent vectors.

2.1 Definitions intrinsically excluding vectors tangent to boundary

We first tried the first option and considered these candidates for definitions which we either came up with ourselves or found in the literature:

Definition 2.1.1 (Our idea inspired by Definition 2.0.1)**.** Let M be a smooth manifold with corners. Let $p \in M$. Then $v \in T_pM$ is inward pointing if and only if there is an $\varepsilon > 0$ and a smooth curve $\gamma\colon [0, \varepsilon) \to M$ such that $\gamma(0) = p$, $\gamma'(0) = v$ and $\gamma((0, \varepsilon)) \subseteq \operatorname{int}(M)$.

Unfortunately, this definition doesn't exclude vectors tangent to the boundary:

Example 2.1.2. Take M to be the upper half plane $\mathcal{H}$ in $\mathbb{R}^2$ let p be $(0, 0)$ and let $\gamma\colon [0, 1) \to \mathcal{H}, x \mapsto x^2$. Then $\gamma'(0) = (1, 0)$ and thus by this definition $(1, 0)$ would be inward pointing even though it is clearly tangent to the boundary. See Figure 2.1 for an illustration of this example.

Then we found this definition by Melrose which he however formulates for vector fields instead of individual vectors. Here is the version for vectors:

Definition 2.1.3 (adapted from [Mel96] Definition 1.13.10)**.** Let M be a smooth manifold with corners. Let $p \in M$. Then $v \in T_pM$ is inward pointing if and only if $\mathrm{D}_v f(p) \geq 0$ for every non-negative $f \in C^\infty(M)$ that is zero on ∂M.

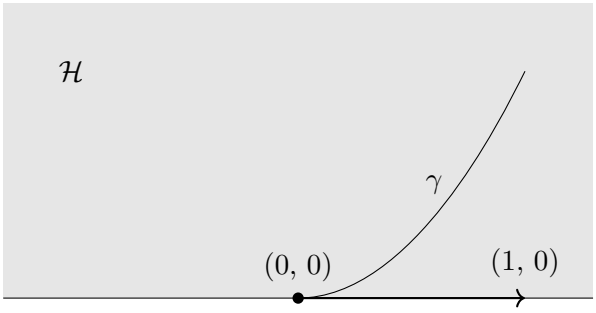


Figure 2.1: Illustration of Example 2.1.2

For some manifolds this definition includes vectors tangent to the boundary:

Example 2.1.4. Take M to be the upper half plane $\mathcal{H}$ in $\mathbb{R}^2$ and let p be $(0,0)$. Let f be any non-negative function in $C^\infty(\mathcal{H})$ that is zero on $\partial\mathcal{H}$. Then $\mathrm{D}_{(0,1)}f(0,0) = 0$. Thus $(0,1)$ is inward pointing by this definition even though it is clearly tangent to the boundary.

It is possible however, that he calls "inward pointing" what we call "realisable". To remedy this we came up with this definition:

Definition 2.1.5 (Our idea inspired by Definition 2.1.3)**.** Let M be a smooth manifold with corners. Let $p \in M$. Then $v \in T_pM$ is inward pointing if and only if there is a non-negative $f \in C^\infty(M)$ that is zero on ∂M such that $\mathrm{D}_v f(p) > 0$.

Unfortunately, both of the above definitions do not work for manifolds with corners:

Example 2.1.6. Take M to be the right upper quadrant $Q_I \subset \mathbb{R}^2$ and let p be $(0,0)$. Let f be any non-negative function in $C^\infty(Q_I)$ that is zero on $\partial\mathcal{Q}_\mathcal{I}$. Then clearly $\mathrm{D}_{(0,1)}f(0,0) = \mathrm{D}_{(1,0)}f(0,0) = 0$. This means that the Jacobian is zero and since f is smooth and thus differentiable at $(0,0)$, this means that $\mathrm{D}_v f(0,0) = 0$ for all vectors v.

So in the case of Definition 2.1.3, this would mean that any vector was inward pointing, and in the case of Definition 2.1.5, no vector would be inward pointing.

This definition however does give the correct notion for smooth vector fields which was perhaps what Melrose intended.

Lastly, here is another popular definition that falls into this category:

Definition 2.1.7 (E.g. [Joy09] Definition 2.2)**.** Let M be a smooth manifold with corners. Let $p \in M$ be a corner point of codimension k, i.e. let there be a chart (U, φ) with $U \subseteq \mathbb{R}^n_k = [0,\infty)^k \times \mathbb{R}^{n-k}$ such that $0 \in U$ and $\varphi(U) = p$. Then a vector $v \in T_pM$ is inward pointing if and only if it is in $\mathrm{d}\varphi_0((0,\infty)^k \times \mathbb{R}^{n-k})$ where $\mathrm{d}\varphi_0 \colon \mathbb{R}^n = T_0\mathbb{R}^n_k \to T_pM$.

This definition is of course correct, however it depends on a choice of chart. In `Mathlib` definitions are generally preferred to not depend on choices. Additionally, it only work for models with corners of the form $\mathbb{R}^n_k$. For `Mathlib` we would need a more general version.

Why exactly? Reference?

2.2 Definitions through realisability

After failing to find a usable definition intrinsically excluding vectors, we moved on to defining realisable vectors. Where as mentioned above, we want a realisable vector to be one that is either inward pointing or tangent to the boundary.

Firstly, we could only find the following way to define inward pointing vectors from realisable ones:

Definition 2.2.1 (Inward Pointing, [Amr21])**.** Let M be a smooth manifold with corners. Let $p \in M$. Then $v \in T_pM$ is inward pointing if and only if it is in the interior of the set of realisable vectors.

Now we can move on to finding a definition of realisability.
We first note conditions for such a definition:

Remark 2.2.2. A sensible definition of realisability needs to fulfil these two conditions:

(i) The definition needs to be preserved by local diffeomorphisms (in particular it should be preserved by smooth charts).

(ii) The definition should reduce to a sensible notion on the model space that is easily verifiable in the cases of the typical examples (e.g. upper half space in $\mathbb{R}^n$).

Let us first discuss what this "sensible notion" in Remark (ii) should be.
We first need the following definition:

Definition 2.2.3 (Contingent Tangent Cone, [AF90] after Definition 4.1.1)**.** Let E be a normed vector space, $K \subset E$ and $x \in \overline{E}$. Then the *contingent tangent cone* to K at x is the set of all $v \in E$ such that there is a sequence of scalars $c\colon \mathbb{N} \to \mathbb{R}_{\geq}$ and a sequence of vectors $d\colon \mathbb{N} \to E$ such that

(i) $\lim_{n\to\infty} d_n = 0$,

(ii) $x + d_n \in K$ for all $n \in \mathbb{N}$, and

(iii) $\lim_{n\to\infty} c_n \cdot d_n = v$.

We denote this tangent cone by $T_K(x)$.

In `Mathlib` this is referred to as `posTangentConeAt`. In the following we will only consider contingent tangent cones to convex sets for which the contingent tangent cone agrees with other notions of tangent cones such as the Clarke tangent cone ([AF90] Proposition 4.2.1). Thus we will refer to it simply as the tangent cone.

This captures exactly our notion of realisable: intuitively, the tangent cone $T_K(x)$ is composed of all vectors at x that are either inward pointing or tangent to K. You can see examples of tangent cones in Figure 2.2.

So we can take the "sensible notion" in Remark (ii) to mean that a vector v should be inward pointing to a point x in the model space if and only if v is in the tangent cone at the point x to the model space.

Now let us consider possibilities for the definition of realisability. The first was already provided to us in Definition 2.0.1:

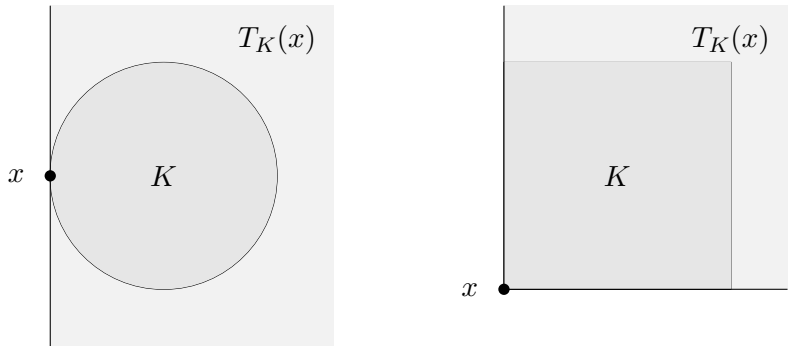


Figure 2.2: Two examples of tangent cones

Definition 2.2.4 ([Lee06] p. 136)**.** Let M be a smooth manifold with corners. Let $p \in M$. Then $v \in T_pM$ is *realisable* if and only if there is an $\varepsilon > 0$ and a smooth curve $\gamma\colon [0, \varepsilon) \to M$ such that $\gamma(0) = p$ and $\gamma'(0) = v$.

It is easy to show that this definition fulfils Remark (i). It also fulfils Remark (ii) but this is surprisingly hard to show for general model spaces H. Taking a vector tangent to the boundary, it is hard to describe a method for picking a smooth curve $\gamma\colon [0, \varepsilon) \to H$ such that $\gamma(0) = p$ and $\gamma'(0) = v$.

This is considerably easier to prove for continuous curves ([Rio18]) or even for curves $\gamma\colon [0, \varepsilon) \to H$ that are just differentiable at 0 ([RW98]). Requiring only minimal conditions for γ eases defining such functions in Lean. So we arrive at the following definition of realisability based on [RW98]:

Definition 2.2.5 (Realisability)**.** Let M be a smooth manifold with corners. Let $p \in M$. Then $v \in T_pM$ is *realisable* if and only if there is a function $\gamma\colon [0, \infty) \to M$ that is differentiable at 0 and fulfils $\gamma(0) = p$ and $\gamma'(0) = v$.

In a normed space this definition reduces to the following which we will call by the name derivability instead of realisability in accordance with [RW98].

Definition 2.2.6 (Derivability)**.** Let E be a normed space over $\mathbb{R}$, let $C \subseteq E$ be a set and let $p \in E$ be a point. We call a vector v *derivable* in C at p if there is and $\varepsilon > 0$ and a function $\gamma\colon [0, \varepsilon) \to E$ that is differentiable at 0 with $\gamma([0, \varepsilon)) \subseteq C$, $\varepsilon(0) = p$ and $\varepsilon'(0) = v$.

The set of all derivable vectors in C at p is called the *derivable cone* to C at p.

2.3 Preservation under charts

Here I should write down the proof that realisability and inward pointing are properties preserved by charts.

2.4 Equivalence with tangent cone on model space

We now need to show that this definition of realisability does indeed equal our desired notion of being in the tangent cone.

For this section let E be a normed space over $\mathbb{R}$, let $C \subseteq E$ be a set and let $p \in E$ be a point.

First we need to show the following lemma:

Lemma 2.4.1. *Let C be convex. Then the derivable cone to C at p is a convex cone.*

The next lemma we need is the following for which the proof is inspired by [Rio18]:

Lemma 2.4.2. *Let $p \in C$. Then the derivable cone is closed in E.*

Now we need yet another cone:

Definition 2.4.3. The set of all vectors v such that there is an $r > 0$ such that $r \cdot v + p \in C$ is called the *feasible cone* to C at p.

Again we get the following lemma:

Lemma 2.4.4. *Let C be convex. Then the feasible cone to C at p is a convex cone.*

Now we can show our first equality of cones:

Lemma 2.4.5. *Let C be convex and $p \in C$. Then the tangent cone to C at p agrees with the closure of the feasible cone to C at p.*

Now we just need to show the following:

Lemma 2.4.6. *Let C be convex and $p \in C$. Then the derivable cone to C at p equals the closure of the feasible cone to C at p.*

Corollary 2.4.7. *Let C be convex and $p \in C$. Then the derivable cone and the tangent to C at p agree.*

Lemma 2.4.8. *Let C be convex and $p \in C$. Then any vector v in the tangent cone to C at p is derivable within C at p.*

Here I should show that this is equivalent to the tangent cone for convex sets.

2.5 Integral Curves

To prove the existence of integral curve, one needs to solve an ODE in the model space. The typical way to do this is via the *Picard-Lindelöf* theorem (also called the *Cauchy-Lipschitz* theorem). This and other similar theorems about ODE are typically formulated for open sets. Since we work with vector fields defined on sets that need not be open, we need to adapt. In a lot of differential geometry textbooks, continuous differentiability and similar notions in not-open sets are defined to mean that there exist local extensions to open neighbourhoods that have this property. However, `Mathlib` has chosen to not do this. So, if we wanted to extend our functions, we would need to have a theorem that permits this. That theorem is the *Whitney extension theorem*. Unfortunately, the proof of this theorem is quite long and thus did not appear to be in scope of this thesis. Additionally, it seems that `Mathlib` has found a lot of beautiful general statements by avoiding this theorem and instead develop a theory that also works for not-open sets.

So this is the route that we also chose to go for. Thus we also needed a version of the Picard-Lindelöf theorem for our use case.

We first collect some conditions for our theorem:

expand on this

discuss this in prereq-uisites sec-tion

cite Gouezels paper

say that this is all based on the mathlib implemen-tation

Definition 2.5.1. Let E be a normed commutative group under addition and a normed space over $\mathbb{R}$. Let $s \subseteq E$ a set, $t_{\min}, t_{\max} \in \mathbb{R}$, $f\colon [t_{\min}, t_{\max}] \times s \to E$ a function, $x_0 \in E$ and $a, r, L, K \in \mathbb{R}_{\geq 0}$. Then we say that f is *Picard-Lindelöf* with respect to x_0, a, r, L, K if the following conditions hold:

(i) *Lipschitz condition*: for all $t \in [t_{\min}, t_{\max}]$, the function $f_t\colon s \to E, x \mapsto f((t,x))$ is Lipschitz on $s \cap D_a(x_0)$ with Lipschitz constant K.

(ii) *Continuity*: for all $x \in s \cap D_a(x_0)$, the function $f^x\colon [t_{\min}, t_{\max}] \to E, t \mapsto f((t,x))$ is continuous.

(iii) *Norm condition*: for all $t \in [t_{\min}, t_{\max}]$ and $x \in s \cap D_a(x_0)$, $\|f((t,x))\| \leq L$.

(iv) *Time condition*: We have $L(t_{\max} - t_{\min}) \leq a - r$.

(v) *Set conditions*: The set s is convex and closed.

(vi) *Direction of vector field*: for all $v \in E$ if there is a $t \in [t_{\min}, t_{\max}]$ such that $\|v - f((t, x_0))\| \leq Ka$, then v is inward pointing with respect to s at x_0.

(vii) *Shape condition:* for all vectors $v \in E$ that are derivable within s at x_0 and fulfil $\|v\| \leq a$, we get $x_0 + v \in s$.

For the rest of the section let us fix a normed commutative additive group E which is also a normed space over $\mathbb{R}$ and a complete space. Additionally fix $t_{\min} \leq t_{\max}$ in $\mathbb{R}$, $s \subseteq E$, a vector field $f\colon [t_{\min}, t_{\max}] \times s \to E$, $x_0 \in s$ and $a, r, L, K \in \mathbb{R}_{\geq}$.

The theorem that we want to prove is the following:

Lemma 2.5.2. *Assume that f is Picard-Lindelöf with respect to x_0, a, r, L, K. The there is a differentiable function $\alpha\colon [t_{min}, t_{max}] \to E$ such that for all $t \in [t_{min}, t_{max}]$ it holds that $\alpha'(t) = f((t, \alpha(t)))$ and $\alpha(t) \in s$.*

Like in the proof of the usual Picard-Lindelöf theorem, we prove this by defining a contraction on a function space and then showing that the fixed point fulfils the conditions required in the theorem.

So let us first define the function space:

Definition 2.5.3. We define $\mathcal{M}$ to be the space of functions $g\colon [t_{\min}, t_{\max}] \to E$ such that

(i) g is Lipschitz continuous with constant L,

(ii) $g(t_{\min}) \in D_r(x_0)$, and

(iii) $g([t_{\min}, t_{\max}]) \subseteq s$.

Since we later want to apply the fact that every contraction in a complete space has a fixed point, we first need to show that this space is complete.

Lemma 2.5.4. *If s is closed, then $\mathcal{M}$ is a complete space with respect to the supremum norm $\|\cdot\|_\infty$.*

Proof. Since $\mathcal{M}$ is a subspace of the complete space $C([t_{\min}, t_{\max}], E)$ it suffices to show that it is closed. We can express $\mathcal{M}$ as the intersection of the following three sets:

$$\{g \in C([t_{\min}, t_{\max}], E) \mid g \text{ is Lipschitz with constant } L\}$$
$$\{g \in C([t_{\min}, t_{\max}], E) \mid \|g(t_{\min}) - x_0\| \leq r\}$$
$$\{g \in C([t_{\min}, t_{\max}], E) \mid g([t_{\min}, t_{\max}]) \subseteq s\}$$

In any metric space, the set of function that a Lipschitz with a specific constant is closed. Thus the first set is closed. The second set is closed since the function $g \mapsto \|g(t_{\min}) - x_0\|$ is continuous. And the last set is closed because s is closed.

Thus, as the intersection of these three sets, $\mathcal{M}$ is also closed.

□

Let us now define the relevant contraction:

Definition 2.5.5. We define the function $C\colon \mathcal{M} \to ([t_{\min}, t_{\max}] \to E)$ as

$$C(g)\colon [t_{\min}, t_{\max}] \to E$$
$$t \mapsto x_0 + \int_{t_{\min}}^{t} f(x, g(x))\ \mathrm{d}x$$

A priori, we do not know that this function again maps to $\mathcal{M}$. So let us prove this.

We first need some helper lemmas:

Lemma 2.5.6. *Let $g\colon [t_{min}, t_{max}] \to E$ such that g is Lipschitz continuous with constant L and $f(t_{min}) \in D_r(x_0)$. Assume additionally that $L(t_{max} - t_{min}) \leq a - r$. Then for all $t \in [t_{min}, t_{max}]$, we get $g(t) \in D_a(x_0)$. In particular, this lemma holds for all $g \in \mathcal{M}$.*

Proof. Take any $t \in [t_{\min}, t_{\max}]$. Then

$$\begin{aligned}\|g(t) - x_0\| &\leq \|g(t) - g(t_{\min})\| + \|g(t_{\min}) - x_0\| \\ &\overset{(1)}{\leq} \|t - t_{\min}\| + r \\ &\leq \|t_{\max} - t_{\min}\| + r \\ &\leq a - r + r = a\end{aligned}$$

where in (2) we used both of the conditions imposed on g. □

Lemma 2.5.7. *Let f be Picard-Lindelöf with respect to x_0, a, r, L, K and let $g \in \mathcal{M}$. Then the function $[t_{min}, t_{max}] \to E, t \mapsto f((t, g(t)))$ is continuous.*

Proof. Using the above lemma and the fact that g is continuous on s, it suffices to show that f is continuous on $[t_{\min}, t_{\max}] \times (s \cap D_a(x_0))$. This then follows from conditions (i) and (ii) of f being Picard-Lindelöf. □

Lemma 2.5.8. *Let f be Picard-Lindelöf with respect to x_0, a, r, L, K and let $g \in \mathcal{M}$. Then $C(g)$ is Lipschitz continuous with constant L.*

Proof. Take $t_1, t_2 \in [t_{\min}, t_{\max}]$. We assume that $t_2 \leq t_1$ Then we get

$$\begin{aligned}
&\|C(f)(t_1) - C(f)(t_2)\| \\
=&\left\| x_0 + \int_{t_{\min}}^{t_1} f(x, g(x)) \, \mathrm{d}x - (x_0 + \int_{t_{\min}}^{t_2} f(x, g(x)) \, \mathrm{d}x) \right\| \\
\overset{(1)}{=}&\left\| \int_{t_2}^{t_1} f(x, g(x)) \, \mathrm{d}x \right\| \\
\leq&\int_{t_2}^{t_1} \|f(x, g(x))\| \, \mathrm{d}x \\
\overset{(2)}{\leq}&\int_{t_2}^{t_1} L \, \mathrm{d}x \\
=&L(t_1 - t_2)
\end{aligned}$$

where in (1) we used that the function $x \mapsto f(x, g(x))$ is continuous by Lemma 2.5.7 and thus integrable. In (2), we used Lemma 2.5.7 and condition (i) of f being Picard-Lindelöf. □

Now we want to show that the range of $C(f)$ lies in s.

Lemma 2.5.9. *Let f be Picard-Lindelöf with respect to x_0, a, r, L, K and let $g \in \mathcal{M}$. Then $C(g)([t_{min}, t_{max}]) \subseteq s$.*

Proof. Let $t \in [t_{\min}, t_{\max}]$. For $t = t_{\min}$ this is clear. So let us assume $t > t_{\min}$. We want to apply condition (vii) of f being derivable. To do that we need to show that $C(f)(t) - x_0$ is derivable within s at x_0 and $\|C(f)(t)\| \leq a$. The second condition holds by Lemma 2.5.6 since $C(f)(t_{\min}) = x_0 \in D_r(x_0)$ and $C(f)$ is Lipschitz with constant L by Lemma 2.5.8.

It is left to show that $C(f)(t) - x_0$ is derivable within a at x_0. Since the derivable cone is a cone, this is equivalent to showing that $(t - t_{\min})^{-1}(C(f)(t) - x_0) = ⨍_{t_{\min}}^{t} f(x, g(x)) \, \mathrm{d}x$ is derivable.

Since the derivable cone is closed and convex and g is integrable, it suffices to show that for all $x \in [t_{\min}, t]$, $f(x, g(x))$ is in the derivable cone to s at x. We show the stronger statement that $f(x, g(x))$ is inward pointing. We want to apply condition (vi) of f being Picard-Lindelöf. To do that we show that $\|f(x, g(x)) - f(x, x_0)\| \leq Ka$. We get:

$$\begin{aligned}
\|f(x, g(x)) - f(x, x_0)\| &\overset{(1)}{\leq} K \, \|g(x) - x_0\| \\
&\overset{(2)}{\leq} Ka
\end{aligned}$$

where in (1) we used that f is Lipschitz continuous in the second variable with constant K by property (i) of f being Picard-Lindelöf. In step (2) we used Lemma 2.5.6. This finishes the proof. □

Corollary 2.5.10. *Let f be Picard-Lindelöf with respect to x_0, a, r, L, K and let $g \in \mathcal{M}$. Then $C(g) \in \mathcal{M}$.*

The above corollary means that we can regard our function C defined in Definition 2.5.5 as a function $C\colon \mathcal{M} \to \mathcal{M}$.

We show that for some n, C^n is a contraction. This will allow us to obtain a fixed point of C^n which is then also a fixed point of C.

We first show the following lemma:

Lemma 2.5.11. *Let f be Picard-Lindelöf with respect to x_0, a, r, L, K, $g_1, g_2 \in \mathcal{M}$ and $n \in \mathbb{N}$. Then*

$$\|C^n(g_1) - C^n(g_2)\|_\infty \leq \frac{(K(t_{max} - t_{min}))^n}{n!} \|g_1 - g_2\|_\infty .$$

Proof. We show by induction on n that for all $t \in [t_{\min}, t_{\max}]$,

$$\|C^n(g_1)(t) - C^n(g_2)(t)\| \leq \frac{(K(t - t_{\min}))^n}{n!} \|g_1 - g_2\|_\infty .$$

For $n = 0$, the inequality reduces to $\|g_1(t) - g_2(t)\| \leq \|g_1 - g_2\|_\infty$ which is clearly true.

Let us now assume that we have shown the inequality for n. We need to show it for $n + 1$.

$$\begin{aligned}
&\left\|C^{n+1}(g_1)(t) - C^{n+1}(g_2)(t)\right\| \\
&= \|(C(C^n(g_1)))(t) - (C(C^n(g_2)))(t)\| \\
&= \left\| \int_{t_{\min}}^{t} f(x, (C^n(g_1))(x)) - f(x, (C^n(g_2))(x)) \, \mathrm{d}x \right\| \\
&\leq \int_{t_{\min}}^{t} \|f(x, (C^n(g_1))(x)) - f(x, (C^n(g_2))(x))\| \, \mathrm{d}x \\
&\overset{(1)}{\leq} \int_{t_{\min}}^{t} K \, \|(C^n(g_1))(x) - (C^n(g_2))(x)\| \, \mathrm{d}x \\
&\overset{(2)}{\leq} \int_{t_{\min}}^{t} K \frac{(K(x - t_{\min}))^n}{n!} \|g_1 - g_2\|_\infty \, \mathrm{d}x \\
&= \frac{(K(t - t_{\min}))^{n+1}}{(n+1)!} \|g_1 - g_2\|_\infty
\end{aligned}$$

In step (1) we used that f is Lipschitz continuous with constant K in the second variable by property (i) of f being Picard-Lindelöf. In step (2) we applied the induction hypothesis. □

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